

Big Data & Hadoop

Hadoop distributes storage and processing across a cluster of computers. HDFS divides files into blocks and replicates them across nodes. MapReduce performs work near the stored data, then combines the results.

Learning objectives

1. Explain why large datasets benefit from distributed processing.
2. Describe the role of replication in fault tolerance.
3. Compare the Map and Reduce stages.

Try it

Sketch how four machines could share eight data blocks while preserving a second copy of each block.